Chapter 1

Rate-Distortion Theory for Caching

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1.1 Introduction

Caching is an appealing solution to reduce network congestion, which is mostly due to media traffic. Indeed, media streaming is one of the main applications of caching, especially since video and music files are resource-demanding, have stable contents, and are usually requested by several users simultaneously. Video and music files often encode details that human auditory or visual system cannot distinguish, and, therefore, most users require only a lossy version of them. In fact, the required level of detail often depends on the user's device. For example, if a song is to be played on a simple mobile phone, the distortions created by the phone's audio equipment (e.g., loudspeakers) render a high-resolution music file useless as opposed to the case of a high-end surround system. Similarly, different video qualities are desired depending on the replay type (HD or SD) and the screen's pixel size. In a setup with multiple users, the desired quality of video or music files thus often vary across different users and files.

In an information-theoretic framework, requirements on quality and resolution of files are typically phrased in terms of distortion measures and maximum allowed distortions. An example of a distortion measure for digital sources often considered in the information theory literature is the *Hamming distortion*, which counts the number of pixels or bits where the reconstruction differs from the original source. For continuous sources, the *squared-error distortion* measure is frequently used, which sums the squares of the differences between any sample and its reconstruction. We are interested in understanding the trade-off between the rate at which a file is transmitted and the minimum distortion with which it can be reconstructed. A *rate-distortion (RD)* function characterizes the smallest compression rate that is needed in order to reproduce a file satisfying the maximum allowed distortion criteria (under a given distortion measure).

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In this chapter, we consider a single-server K-user centralized caching setup with a library of correlated files. In the first communication phase, *the placement phase*, the library prefetches and stores contents in the users' cache memories. Later, each user demands one specific file from the library and the demands are revealed to the server and all the users in the system. During the second communication phase, *the delivery phase*, the server sends a common delivery message that is observed by all the users in the system and allows the users to reconstruct, using the contents in their local cache memory, their requested file with the desired distortion. The main focus of this chapter is on the minimum rate of this delivery message given a set of distortion criteria and cache sizes at the users. We call this quantity the *rate-distortion-memory (RDM) function* of the system.

The RDM problem for the single-server $K \geq 1$-user setup was introduced by Timo *et al.* in [1, 2]. In their setup (reproduced here in Section 1.3) the distortion measure can differ over files and users. This assumption is motivated by the different types of the source files (audio or video files) and by the different kinds of user terminals (mobile phones or high-end surround systems). The RDM function for this model is still widely open, why in this chapter we focus on interesting special cases.

The first special case is the single-user model (Section 1.4), for which Timo *et. al* [1, 2] determined the optimal RD function and the optimal placement and delivery strategies for arbitrary sources and distortion measures (under some technical conditions that we will explain). The single-user setup with bivariate Gaussian sources and squared-error distortion measures was also treated by Op't Veld and Gastpar in [3, 4]. Both lines of work exploit the connection of the single-user RD caching problem to the Gray-Wyner source coding network and draw connections to known common information (CI) measures such as lossy Wyner [5–7] and Gács-Körner [7–10] CIs.

Slightly more involved is the two-user case, which we treat in Section 1.5. Here, a complete solution seems out of reach and the literature has focused on various special cases. The work in [1] finds lower and upper bounds for the case where only one of the two users has a cache memory, and [11, 12] focus on lossless caching and propose two-step strategies in which source coding and coded caching are separated.

Finally, due to the complexity of the problem, for the general K-user scenario the literature has focused on *uncorrelated* Gaussian sources and mean squared-error distortion. As we explain in Section (1.6) of this chapter, under this assumption, it is natural to compress each file individually into different layers, and upon request deliver each user the number of layers allowing it to attain the desired distortion. Scalable source coding [13–15] is typically applied as a framework for this approach. Yang and Gündüz [16, 17], Hassazadeh *et. al* [18], and Zewaili and Yener [19] considered such a scalable source-coding approach for uncorrelated Gaussian sources (files) under mean-squared error distortion measures. More specifically, Yang and Gündüz [16, 17] proposed practical cache allocation algorithms to identify what percentage of the cache memory each user should allocate to each layer. In their approach, placement and delivery are performed separately for each layer. This per-layer approach was proved by Zewaili and Yener [19] to be generally suboptimal under uncoded placement and linear delivery, but optimal under these assumptions

when one can optimize the assignment of the cache sizes to the different users subject to a total cache memory budget. (The problem of cache assignment was first studied in [20, 21] for the standard coded caching setup over a noisy broadcast channel.) Hassanzadeh *et. al* [22] considered a different global optimization problem across users, where the maximum permissible distortion levels are not fixed for each user and file. Instead, in their approach the goal is to minimize the *average* mean squared-error distortion across all the K users and all files (which are requested according to a given file popularity distribution).

Related, but slightly beyond the scope of this chapter are also the "caching for computing" framework of [23, 24] and the RD formulation of the caching problem over noisy broadcast channels (BC). Wang *et al.* [23, 24] formulated the "caching for computing" problem that is more resemblant of traditional information-theoretic source coding with side-information. In their scenario a user does not emit a single request to indicate the file that it wishes to download, but it makes a new request for each symbol. The request sequence is then modelled as the side-information sequence of a standard source-coding problem. Since the focus of [23] is on *lossless compression*, we will not treat it in detail in this chapter. It is however obvious that similar results can also be obtained for a RD formulation of their problem.

So far, delivery communication was assumed to take place over a noise-less rate-limited link. Timo *et al.* introduced a variation of the coded caching problem where delivery communication is over a noisy BC [25, 26]. They showed at hand of erasure BCs that joint source-channel coding (also termed *joint cache-channel coding* in this context) can greatly improve the performance of lossless caching systems when delivery is over a noisy BC. Similar results were also obtained for other channels such as the K-user Gaussian BC [20, 21]. The RD caching problem with correlated files and delivery over noisy BCs still remains an open research problem. Simple results can be obtained by appropriately combining previous results on the lossless caching over noisy BCs with the RD results presented in this chapter. The lossless counterpart of the RD caching problem over a Gaussian BC has been studied by Yang *et al.* [27].

1.2 Rate-Distortion Functions

We provide a quick overview of the basic quantities in RD theory used in this chapter: the single-source RD function, the joint RD function and the conditional RD function. More details can be found in standard textbooks, see [28, 29].

Given a source V following some distribution P_V over an alphabet $\mathscr{V}$, a reconstruction alphabet $\hat{\mathscr{V}}$, a non-negative and bounded per-letter distortion measure $\delta\colon \hat{\mathscr{V}} \times \mathscr{V} \to \mathbb{R}^+$ (where the latter set denotes the nonnegative reals), and a nonnegative distortion $D \geq 0$, the *single-source RD function* is defined as

$$R_V(D) := \min I(V;\hat{V}), \tag{1.1}$$

where the minimum is over all auxiliary random variables $\hat{V}$ over the set $\hat{\mathscr{V}}$ jointly distributed with V and satisfying:

$$\mathbb{E}[\delta(\hat{V},V)] \leq D. \tag{1.2}$$

The single-source RD function determines the minimum number of bits *per source symbol* required to describe an *independent and identically distributed (i.i.d.)* source sequence $V^n := (V_1, \ldots, V_n)$ distributed according to P_V such that a decoder that has access to this description can produce a reconstruction $\hat{V}^n \in \hat{\mathscr{V}}$ that matches the original source up to average per-letter distortion D in the asymptotic limit as the blocklength n tends to infinity, i.e., so that

$$\overline{\lim_{n\to\infty}} \frac{1}{n} \sum_{i=1}^{n} \delta(\hat{V}_i, V_i) \leq D. \tag{1.3}$$

The joint RD function has an analogous meaning when the single source V is replaced by a tuple $(V_1, \ldots, V_t) \sim P_{V_1 \cdots V_t}$. In particular, it determines the minimum number of bits *per source tuple* required to *jointly describe* the i.i.d. source sequences $V_1^n, \ldots, V_t^n$, for $V_j^n := (V_{j,1}, \ldots V_{j,n})$ and $(V_{1,i}, \ldots, V_{t,i}) \sim P_{V_1 \cdots V_t}$, with a common message index J, such that a decoder that has access to J can produce reconstructions $\hat{V}_1^n, \ldots, \hat{V}_t^n$ that individually match the original sources up to average per-letter distortion $D_1, \ldots, D_t$ in the asymptotic limit as the blocklength n tends to infinity. Formally, the joint RD function is defined as

$$R_{V_1 \cdots V_L}(D) := \min I(V_1, \ldots, V_t; \hat{V}_1, \ldots, \hat{V}_t), \tag{1.4}$$

where the minimum is over all auxiliary random variables $\hat{V}_1, \ldots, \hat{V}_t$ over the given reconstruction sets $\hat{\mathscr{V}}_1, \ldots, \hat{\mathscr{V}}_t$ jointly distributed with $V_1, \ldots, V_t$ and satisfying:

$$\mathbb{E}[\delta_j(\hat{V}_j, V_j)] \leq D_j, \qquad j \in \{1, \ldots, t\}. \tag{1.5}$$

Finally, the *conditional RD function* has a similar operational meaning as the single-file RD function, but when the encoder and the decoder (who produces the reconstruction) have access to a random sequence $S^n := (S_1, \ldots, S_n)$ where for each $i \in \{1, \ldots, n\}$ the pair (S_i, V_i) follows a given joint distribution P_{SV}. Formally, the conditional RD function is defined as

$$R_{V|S}(D) := \min I(V; \hat{V}|S), \tag{1.6}$$

where the minimum is over all auxiliary random variables $\hat{V}$ over the given reconstruction set $\hat{\mathscr{V}}$ jointly distributed with the pair (S, V) and satisfying:

$$\mathbb{E}[\delta(\hat{V}, V)] \leq D. \tag{1.7}$$

1.3 Description of the System Model

We assume throughout this chapter that the library consists of $(L \geq 1)$ different source files, denoted by

$$W_1^n, \ldots, W_L^n.$$

We further assume that the ℓ-th source file, W_ℓ^n, where $\ell \in \mathscr{L} := \{1, 2, \ldots, L\}$, consists of n symbols chosen from a discrete and finite alphabet[4] $\mathscr{W}_\ell$. That is, we can write the ℓ-th source file as

$$W_\ell^n := (W_{\ell,1}, \ldots, W_{\ell,n}),$$

[4]We will drop the finite alphabet assumption for some Gaussian source examples.

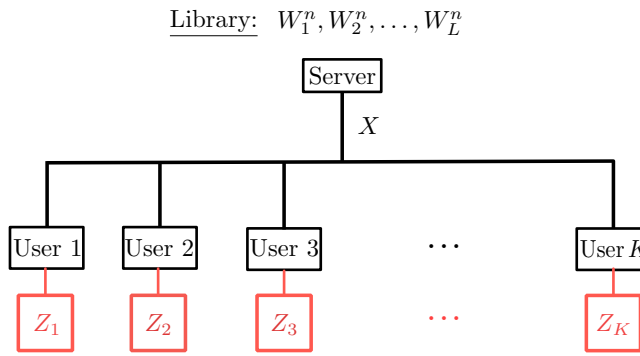

Figure 1.1: Illustration of the general model with a single server and K cache-aided users.

for $W_{\ell,1}, \ldots, W_{\ell,n} \in \mathscr{W}_\ell$. The number of source symbols n (also called the blocklength) will be allowed to grow without bound so as to enable an information-theoretic analysis. Thus, we are interested in libraries consisting of a fixed number of large source files.

We further assume that the library is randomly generated by a discrete memoryless source (DMS), i.e., $W_1^n, \ldots, W_L^n$ is the sequence of L-tuples $\{(W_{1,i}, \ldots, W_{L,i})\}_{i=1}^n$ and is i.i.d. according to a given, but arbitrary, joint distribution $P_{W_1 \cdots W_L}$ defined on $\mathscr{W}_1 \times \cdots \times \mathscr{W}_L$:

$$(W_{1,i}, \ldots, W_{L,i}) \sim P_{W_1 \cdots W_L}, \qquad i \in \{1, \ldots, n\}. \tag{1.8}$$

This assumption is quite common in the multi-terminal RD theory literature, as it admits rigorous proofs and gives some insight to more complicated models. Although it is somewhat restrictive, some important transformations (e.g. Burrows-Wheeler) are known to emit almost memoryless processes [30, 31].

We consider a communication system that consists of a single server and K users, which we enumerate from 1 to K (see Figure 1.1). The server has access to the entire library $W_1^n, \ldots, W_L^n$ and each user k is interested in learning one of these files. It is however à priori unknown which files the users are interested in. User k has access to a local cache memory whose content is denoted by Z_k and that can fit up to $n \cdot M_k$ bits. We thus say that the cache memory has a capacity of M_k bits per source symbol.

As in the previous chapters of this book, we assume that the server can prefetch arbitrary contents in all cache memories during an idle-network period when the users have not yet decided (or communicated) which files they wish to download. In this chapter we mostly concentrate on the so called *centralized caching scenario* where the server centrally decides on all cache contents, as opposed to a *decentral-*

ized caching scenario where each user locally decides on how to fill its own cache memory. The cache memories $Z_1, \dots, Z_K$ are thus filled according to

$$Z_k = \phi_k(W_1^n, \dots, W_L^n), \qquad k \in \{1, \dots, K\}, \tag{1.9}$$

for arbitrary placement functions $\phi_1, \dots, \phi_K$ on appropriate domains.

Following the placement phase where the users' cache memories are filled with contents, they communicate to the server and the other users in the system which files they wish to learn.[5] It will be interesting to allow for restricted demand sets at the various users. That means, assume given subsets

$$\mathscr{L}_1, \dots, \mathscr{L}_K \subseteq \mathscr{L}, \tag{1.10}$$

where for each k the set $\mathscr{L}_k$ contains the indices of all files that can be requested by user k. In most of the caching literature, all users can potentially demand all files, i.e., $\mathscr{L}_1 = \cdots = \mathscr{L}_K = \mathscr{L}$. Certain situations, e.g., systems with different kinds of subscriptions, are however more adequately modeled with unequal demand sets $\mathscr{L}_1, \dots, \mathscr{L}_K$. In any case, at the end of the placement phase, each user k produces a demand

$$\ell_k \in \mathscr{L}_k, \tag{1.11}$$

and in the subsequent *delivery* phase the server conveys a message X taking value in a finite index set $\mathscr{X}$ over a noise- and error-free link to all the users. Mathematically speaking, the message X is of the form

$$X = f(W_1^n, \dots, W_L^n, \ell_1, \dots, \ell_K, Z_1, \dots, Z_K), \tag{1.12}$$

for some delivery encoding function f on appropriate domains.

The goal is to design the message X in a way that each user can reconstruct its desired file up to a maximum admissible distortion. In practice, a given user might decide on this maximum admissible distortion and the specific distortion measure at the moment where it requests the file, for example in function of the current channel conditions or on the specific application running on the device. This however makes the system more complicated. For simplicity and to better illustrate the impact of lossy reconstructions on a caching system, we assume throughout this chapter that for each file W_ℓ^n each user k has a fixed maximum admissible distortion level $D_{k,\ell}$ under a given distortion measure $\delta_{k,\ell}^{(n)}$. Moreover, both sets $\{D_{k,\ell}\}$ and $\{\delta_{k,\ell}^{(n)}\}$ are known to the server and all the users in advance, before the cache memories are filled with contents.

Each user k is thus supposed to apply a reconstruction function g_k on appropriate domains to the received message X and its local cache content Z_k so that in the limit of large blocklengths n, the reconstruction

$$\hat{W}_k^n = g_k(X, Z_k, \ell_1, \dots, \ell_K) \tag{1.13}$$

satisfies the distortion constraint

$$\overline{\lim_{n\to\infty}} \, \mathbb{E}\big[\delta_{k,\ell_k}^{(n)}\big(\hat{W}_k^n, \hat{W}_{\ell_k}^n\big)\big] \leq D_{k,\ell_k}, \quad \forall k \in \{1, \dots, K\}. \tag{1.14}$$

[5] It suffices that the server learns the requests, because it can communicate it using $K \log L$ bits to all users, which requires 0 delivery rate.

In this chapter, we consider *separable distortion functions* that take on the form

$$\delta_{k,\ell}^{(n)}(\hat{W}_\ell^n, W_\ell^n) = \frac{1}{n}\sum_{i=1}^{n} \delta_{k,\ell}(\hat{W}_{\ell,i}, W_{\ell,i}) \tag{1.15}$$

for some per-letter distortion function $\delta_{k,\ell}\colon \hat{\mathscr{W}}_\ell \times \mathscr{W}_\ell \to [0,\infty)$. We allow for arbitrary (not only squared-error) per-letter distortion measures $\{\delta_{k,\ell}\}$ subject to the following two conditions:[6]

1. For each source symbol $w_\ell \in \mathscr{W}_\ell$ there exists a reconstruction symbol $\hat{w}_\ell \in \hat{\mathscr{W}}_\ell$ such that $\delta_{k,\ell}(\hat{w}_\ell, w_\ell) = 0$; and
2. there exists a finite $D_{\max} > 0$ such that $\delta_{k,\ell}(\hat{w}_\ell, w_\ell) \le D_{\max}$ for all $w_\ell \in \mathscr{W}_\ell$ and $\hat{w}_\ell \in \hat{\mathscr{W}}_\ell$, and all $k \in \{1,\dots,K\}$ and $\ell \in \mathscr{L}$.

For example, $\delta_{k,\ell}$ can be the *Hamming distortion function* where $\hat{\mathscr{W}}_\ell = \mathscr{W}_\ell$ and for all $\hat{w}_\ell, w_\ell \in \mathscr{W}_\ell$:

$$\delta_{k,\ell}(\hat{w}_\ell, w_\ell) = \begin{cases} 1 & \text{if } \hat{w}_\ell \neq w_\ell \\ 0 & \text{if } \hat{w}_\ell = w_\ell. \end{cases}$$

We will further restrict to a maximum *expected* distortion criterion as is more common both in the general information-theoretic literature and in the caching literature.

Alternatively, one could also consider a maximum *excess* distortion criterion instead of the abovementioned maximum expected distortion criterion, or allow for the more general f-*separable distortion measures*. Under an f-separable distortion measure the n-symbol distortion is calculated as [32]

$$\delta^{(n)}(\hat{w}^n, w^n) := \mathsf{f}_\ell^{-1}\left(\frac{1}{n}\sum_{i=1}^{n} \mathsf{f}_\ell\big(\mathsf{d}_\ell(\hat{w}_i, w_i)\big)\right), \tag{1.16}$$

for some given continuous and strictly increasing function $\mathsf{f}_\ell\colon [0,\infty) \to [0,\infty)$. For example f_ℓ can be chosen to assign appropriate (possibly non-linear) frequency costs to different quantization error events. Results for f-separable distortion measures and maximum excess distortion constraints were presented in [2].

As mentioned previously, in this chapter, we focus on separable distortion measures and the maximum expected distortion criterion.

Definition 1. *Let* $\mathbf{D} := (D_{1,1},\dots,D_{1,L},D_{2,1},\dots,D_{2,L},\dots,D_{K,1},\dots,D_{K,L})$ *and* $\mathbf{M} := (M_1,\dots,M_K)$ *be arbitrary nonnegative allowed distortions and cache sizes at the users, and* $\{\delta_{k,\ell}\}$ *arbitrary per-letter distortion measures. We say that a delivery rate* $R \ge 0$ *is* $(\mathbf{D},\mathbf{M})$-admissible *if for every* $\varepsilon > 0$ *there exists a sufficiently large blocklength* n *and placement, delivery encoding, and reconstruction functions* $\phi_1,\dots,\phi_K, f, g_1,\dots,g_K$ *such that for all* K*-tuples* $(\ell_1,\dots,\ell_K) \in \mathscr{L}_1 \times \cdots \times \mathscr{L}_K$*:*

$$\mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} \delta_{k,\ell_k}(\hat{W}_{k,i}, W_{\ell_k,i})\right] \le D_{k,\ell_k} + \varepsilon, \qquad \forall k \in \{1,\dots,K\},$$

[6] An exception is made for Gaussian sources with squared-error distortion measures where assumption 2. is not valid.

and the cardinality of the delivery message is bounded as

$$\frac{1}{n}\log_2|\mathscr{X}| \leq R+\varepsilon, \quad \forall(\ell_1,\ldots,\ell_K)\in\mathscr{L}_1\times\cdots\times\mathscr{L}_K. \tag{1.17}$$

We thus employ a *worst-case* perspective where the delivery rate needs to suffice to deliver the required information irrespective of the actual demands. Alternatively, one can also focus on the *average delivery rate* in which case the alphabet of the delivery message X can depend on the demand vector

$$\boldsymbol{\ell} := (\ell_1,\ldots,\ell_K). \tag{1.18}$$

Under the *average-case* perspective, we write $\mathscr{X}_{\boldsymbol{\ell}}$ for the alphabet of X under the demand vector $\boldsymbol{\ell}=(\ell_1,\ldots,\ell_K)$, and replace (1.17) by

$$\frac{1}{|\mathscr{L}_1|\cdots|\mathscr{L}_K|}\sum_{\substack{\boldsymbol{\ell}=(\ell_1,\ldots,\ell_K)\\ \in\mathscr{L}_1\times\cdots\times\mathscr{L}_K}}\frac{1}{n}\log_2\left|\mathscr{X}_{\boldsymbol{\ell}}\right| \leq R+\varepsilon, \quad \forall\boldsymbol{\ell}\in\mathscr{L}_1\times\cdots\times\mathscr{L}_K. \tag{1.19}$$

Through a slight change, it is also possible to account for file popularities and to focus on the *expected* delivery rate. Under this perspective, we let $p_{k,\ell}$ denote the probability that user k demands file W_ℓ, and replace Constraint (1.19) by

$$\sum_{\substack{\boldsymbol{\ell}=(\ell_1,\ldots,\ell_K)\\ \in\mathscr{L}_1\times\cdots\times\mathscr{L}_K}}\left(\prod_{k=1}^{K}p_{k,\ell_k}\right)\cdot\frac{1}{n}\log_2\left|\mathscr{X}_{\boldsymbol{\ell}}\right| \leq R+\varepsilon. \tag{1.20}$$

Under any of the three perspectives, we call $\mathbf{M}$ the *cache memory vector* and $\mathbf{D}$ the *distortion constraint vector*. The optimal rate-distortion-memory (RDM) trade-off in the asymptotic regime of infinite blocklengths $n\to\infty$ is characterised by the following function.

Definition 2. *The* worst-case RDM function *is*

$$\mathsf{R}_{\text{worst}}(\mathbf{D},\mathbf{M}) := \inf\left\{R\geq 0 : R \textit{ is } (\mathbf{D},\mathbf{M})\textit{-admissible}\right\}.$$

The average RDM function $\mathsf{R}_{\text{avg}}(\mathbf{D},\mathbf{M})$ *and the* expected RDM function $\mathsf{R}_{\text{exp}}(\mathbf{D},\mathbf{M})$ *are defined analogously.*

Our main focus will be on the worst-case RDM function, but we will also discuss results on the average and expected RDM functions. Some simple properties of the worst-case RDM function $\mathsf{R}_{\text{worst}}(\mathbf{D},\mathbf{M})$ have been proved in [2] for $K=1$ or $K=2$. We first note that the worst-case RDM function $\mathsf{R}_{\text{worst}}(\mathbf{D},\mathbf{M})$ is jointly convex, continuous and non-increasing in each component of the vectors $\mathbf{D}$ and $\mathbf{M}$. Notice further that without cache memories, $\mathbf{M}=(0,0,\ldots,0)$, all information has to be sent in the delivery phase. By the broadcast nature of the delivery channel (every user observes the entire transmission), the worst-case RDM function for the *all-zero* cache memory vector $\mathbf{M}=\mathbf{0}:=(0,0,\ldots,0)$ is:

$$\mathsf{R}_{\text{worst}}(\mathbf{D},\mathbf{0}) = \max_{\substack{(\ell'_1,\ldots,\ell'_K)\\ \in\mathscr{L}_1\times\cdots\times\mathscr{L}_K}}\mathsf{R}_{W_{\ell'_1}\cdots W_{\ell'_K}}(D_{1,\ell'_1},\ldots,D_{K,\ell'_K}),$$

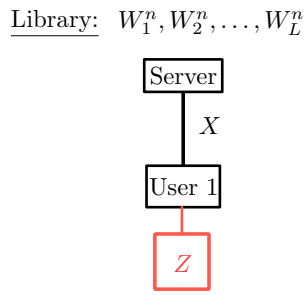

Figure 1.2: Illustration of single-user caching model.

where recall that (see Section 1.2) $\mathsf{R}_{W_{\ell'_1}\cdots W_{\ell'_K}}(D_{1,\ell'_1},\ldots,D_{K,\ell'_K})$ denotes the standard *joint RD function* for the file tuple $W_{\ell'_1},\ldots,W_{\ell'_K} \sim P_{W_{\ell'_1}\cdots W_{\ell'_K}}$.

In the other extreme, when the cache memories are sufficiently large so that all relevant information can be stored at every single user, then no delivery message is required. That means, for any cache memory vector $\mathbf{M} = (M_1,\ldots,M_K)$ and distortion vector $\mathbf{D}$ satisfying:

$$M_k \geq \mathsf{R}_{W_1\cdots W_L}(D_{k,1},\ldots,D_{k,L}), \quad \forall k \in \{1,\ldots,K\}, \tag{1.21}$$

the RDM function vanishes:

$$\mathsf{R}_{\text{worst}}(\mathbf{D},\mathbf{M}) = 0. \tag{1.22}$$

Determining the worst-case RDM function in general is challenging. In fact, even for lossless reconstructions the problem remains unsolved. We will therefore focus on interesting special cases to get some insight on the general problem.

1.4 Single-User Systems

We start by considering a single user ($K = 1$), see Figure 1.2, and without ambiguity drop the user index 1 to simplify notation. That means, we denote the single user's cache content by Z, its size by M, the single-user's request by ℓ, the demand set by $\mathscr{L}$, the per-letter distortion measure for a given file W_ℓ^n by δ_ℓ, and its maximum permissible distortion by D_ℓ.

The results in this section on the worst-case RDM function are based on [1, 2], and the results on the average-case RDM function in the Gaussian case are based on [3, 4].

1.4.1 The RDM functions of the Single-User System

We start by deriving the RDM functions of the single-user caching system, by making use of an operational equivalence between the caching system and the Gray-

Wyner source coding network [33], as well as the tight characterization of the RD region for this problem found by Gray and Wyner in their original manuscript.

1.4.1.1 The Gray-Wyner Source Coding System

The single-user caching system is closely related to the Gray-Wyner source coding network [33] in Figure 1.3. The Gray-Wyner network consists of a single encoder observing the correlated sources $W_1^n, \ldots, W_L^n$ and L decoders $\ell = 1, \ldots, L$, each wishing to reconstruct their corresponding source W_ℓ with a prescribed maximum distortion level $\tilde{D}_\ell$ under a given per-letter distortion measure δ_ℓ. The Gray-Wyner network is comprised of a common rate-R_{c} communication link from the encoder to *all decoders* and L individual communication links of rates $R_1, \ldots, R_L$ from the encoder to each of the decoders. Let $\mathbf{R}_{\mathrm{ind}} := (R_1, \ldots, R_L)$ denote the individual-rates vector and $\tilde{\mathbf{D}} := (\tilde{D}_1, \ldots, \tilde{D}_L)$ the distortions vector. By standard convention, a RD tuple $(R_{\mathrm{c}}, \mathbf{R}_{\mathrm{ind}}, \tilde{\mathbf{D}})$ is called achievable in the Gray-Wyner network if it is possible to find a sequence of encoders and decoders satisfying the rate constraints and the maximum allowed distortions [6, 7]. The set of all achievable RD tuples $(R_{\mathrm{c}}, \mathbf{R}_{\mathrm{ind}}, \tilde{\mathbf{D}})$ for the

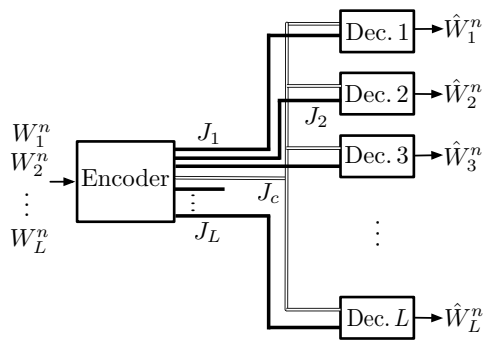

Figure 1.3: Illustration of the Gray-Wyner source coding network with a common message J_{c} of rate R_{c} and individual messages $J_1, \ldots, J_L$ of rates $R_1, \ldots, R_L$

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L-decoder Gray-Wyner network is given by [6, 7]

$$\mathscr{R}_{\mathrm{GW}}(\tilde{\mathbf{D}}) := \bigcup_{P_{U|W_1 \cdots W_L}} \left\{ (R_{\mathrm{c}}, \mathbf{R}_{\mathrm{ind}}) : \begin{array}{lcl} R_{\mathrm{c}} & \geq & I(W_1, \ldots, W_L; U) \\ R_{\ell'} & \geq & \mathsf{R}_{W_{\ell'}|U}(\tilde{D}_{\ell'}) \quad \forall\, \ell' \in \mathscr{L} \end{array} \right\}, \quad (1.23)$$

where recall that (see Section 1.2) $\mathsf{R}_{W_{\ell'}|U}(\tilde{D}_{\ell'})$ is the conditional RD function of file $W_{\ell'}^n$ given a sequence U^n that is jointly i.i.d. with $W_{\ell'}$ according to $P_{UW_{\ell'}}$ and known both to the encoder and the decoder.

The following scheme achieves the RD tuples in $\mathscr{R}_{\mathrm{GW}}(\tilde{\mathbf{D}})$. The encoder quantises the n-tuple source $W_1^n, \ldots, W_L^n$ into a codeword U^n and sends the description using the common index J_{c} to all the decoders. Subsequently, the encoder applies a

separate conditional RD code to each file W_ℓ^n, for $\ell \in \mathscr{L}$, assuming side-information U^n at both the encoder and decoder and for the target distortion $\tilde{D}_\ell$. It finally sends the resulting description $\hat{W}_\ell$ using the private index J_ℓ to Decoder ℓ, which produces $\hat{W}_\ell$ as its reconstruction.

1.4.1.2 An Operational Equivalence and the RDM Function

We now present an observation that is key to solving the single-user RD caching problem of this section. The Gray-Wyner RD problem is equivalent to the single-user caching system, as seen through the following associations.

- The cache memory Z in the caching setup corresponds to the index J_c sent over the common link in the Gray-Wyner source coding setup, and the cache size M to the common rate R_c in the Gray-Wyner setup.
- If the single user in the caching system demands file W_ℓ, then it can be associated with User ℓ in the Gray-Wyner network. Under demand ℓ, the delivery message X in the caching setup then corresponds to the private index J_ℓ in the Gray-Wyner network and the delivery rate $\frac{1}{n}\log|\mathscr{X}|$ corresponds to Gray-Wyner's individual rate R_ℓ.
- Consequently, the worst-case, average, and expected RDM functions of the caching setup are given by the maximum, average, and expected Gray-Wyner individual rates. Applying the expression for the set of achievable RD tuples in (1.23) then yields the following worst-case RDM function:

$$\begin{aligned}
\mathsf{R}_\mathrm{worst}(\mathbf{D},M) &= \min_{\mathbf{R}\colon (M,\mathbf{R})\in\mathscr{R}_\mathrm{GW}(\mathbf{D})} \max_{\ell\in\mathscr{L}} R_\ell && (1.24)\\
&= \min_{U\colon I(W_1,\ldots,W_L;U)\le M} \max_{\ell\in\mathscr{L}} \mathsf{R}_{W_\ell|U}(D_\ell), && (1.25)
\end{aligned}$$

and for the average RDM function:

$$\begin{aligned}
\mathsf{R}_\mathrm{avg}(\mathbf{D},M) &= \min_{\mathbf{R}\colon (M,\mathbf{R})\in\mathscr{R}_\mathrm{GW}(\mathbf{D})} \frac{1}{L}\sum_{\ell\in\mathscr{L}} R_\ell && (1.26)\\
&= \min_{U\colon I(W_1,\ldots,W_L;U)\le M} \frac{1}{L}\sum_{\ell\in\mathscr{L}} \mathsf{R}_{W_\ell|U}(D_\ell). && (1.27)
\end{aligned}$$

The expression for the *expected* RDM function $\mathsf{R}_\mathrm{avg}(\mathbf{D},M)$ is analogous and omitted.

1.4.1.3 Two Examples

Even though (1.25) and (1.27) characterize the worst-case and average RDM functions exactly, an analytical evaluation of these formulas for specific examples can be tricky because it is unclear how to choose the auxiliary random variable U. We present the bounds on the worst-case RDM function $\mathsf{R}_\mathrm{worst}(\mathbf{D},M)$ found in [2] for two canonical examples: the *doubly symmetric binary source* (DSBS) under Hamming distortions with $D_1 = D_2 = 0$ and a centered bivariate Gaussian source under squared-error distortion. For the Gaussian example, we also provide the bound on the average-case RDM function $\mathsf{R}_\mathrm{avg}(\mathbf{D},M)$ found in [4].

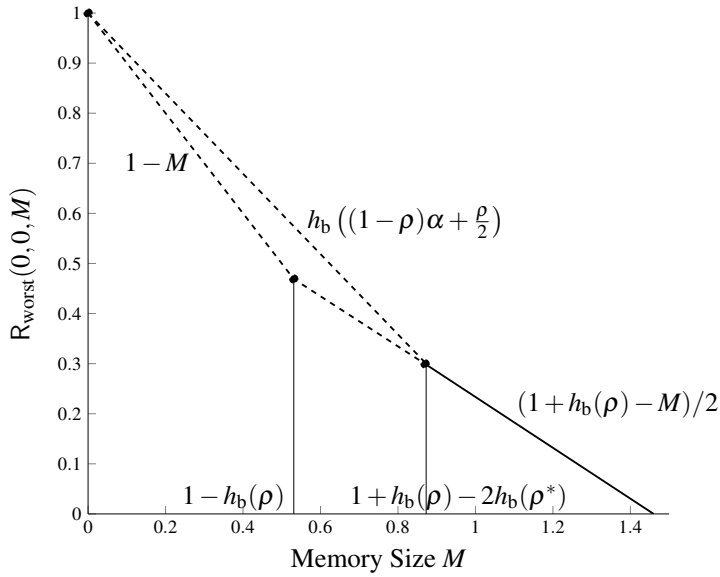

Figure 1.4: Illustration of the upper (achievable) and lower (converse) bounds on the RDM function of DSBS sources with correlation coefficient $\rho = 0.1$.

Example 1 (Doubly-Symmetric Binary Source with Lossless Reconstructions)**.** *Consider a source with only two files (W_1^n, W_2^n) whose symbols are binary and follow the distribution*

$$P_{W_1W_2}(w_1,w_2) = \frac{1}{2}(1-\rho)\cdot \mathbb{1}\{w_1 = w_2\} + \frac{1}{2}\rho \cdot \mathbb{1}\{w_1 \neq w_2\}, \quad w_1, w_2 \in \{0,1\},$$

for a given parameter $\rho \in [0,1/2]$. Consider the same Hamming distortion measure for all files: $\delta_\ell(\hat{w}, w) = \mathbb{1}\{w \neq w\}$, for all $w, \hat{w} \in \{0,1\}$ and $\ell \in \mathscr{L}$. Let h_b denote the binary entropy function:

$$h_b(t) := -t\log t - (1-t)\log(1-t), \quad t \in [0,1], \tag{1.28}$$

and define $\rho^* := \frac{1}{2} - \frac{1}{2}\sqrt{1-2\rho}$.

Under zero distortions $D_1 = D_2 = 0$, depending on the available cache size M, the worst-case RDM function is bounded as follows:

- *If $1 + h_b(\rho) - 2h_b(\rho^*) \leq M \leq 1 + h_b(\rho)$, then*

$$R_{worst}(\mathbf{0}, M) = \frac{1 + h_b(\rho) - M}{2}. \tag{1.29}$$

- *If $0 \leq M < 1 - h_b(\rho)$, then*

$$R_{worst}(\mathbf{0}, M) \geq [1-M]^+. \tag{1.30}$$

- *If* $0 \leq M \leq 1 + h_{\mathrm{b}}(\rho) - 2h_{\mathrm{b}}(\rho^*)$, *then*

$$\frac{1 + h_{\mathrm{b}}(\rho) - M}{2} \leq \mathsf{R}_{\mathrm{worst}}(\mathbf{0}, M) \leq h_{\mathrm{b}}\left((1-\rho)\alpha + \frac{\rho}{2}\right), \tag{1.31}$$

  where $\alpha := h_{\mathrm{b}}^{-1}\left(\frac{1-\rho-M}{1-\rho}\right)$ *and* h_{b}^{-1} *denotes the inverse binary entropy function that takes values in the interval* $[0, 1/2]$.

Above bounds can be proved following similar steps as in the DSBS examples in [33, Sec. 1.5], [34, Sec. III.C] and [23, Ex. 1]. The same steps can also be used to show that the bounds remain valid for the average-case RDM function $\mathsf{R}_{\mathrm{avg}}(\mathbf{D}, M)$ *(which by definition is always smaller than or equal to the worst-case RDM function).*

Bounds (1.29)–(1.31) *are illustrated in Figure 1.4. We use a solid line to indicate the exact worst-case RDM function in the regime where it is known, and dashed lines in the regime where only upper and lower bounds are known.*

In the second example, we consider a bivariate Gaussian source and squared-error distortion measures. For more than two sources, a lower bound on the worst-case RDM-function is presented in [2].

Example 2. *Consider a library with two files* $\{W_1^n, W_2^n\}$ *generated by a centered bivariate Gaussian distribution with covariance matrix*

$$\mathsf{K}_{W_1W_2} = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}, \tag{1.32}$$

for a given correlation coefficient $\rho \in (0,1)$. *Further assume squared-error distortion functions for both files, i.e.,* $\delta_1(\hat{w}, w) = \delta_2(\hat{w}, w) = (\hat{w} - w)^2$, *for all* $w, \hat{w} \in \mathbb{R}$, *and symmetric maximum distortions* $D_1 = D_2 = D \geq 0$. *The joint RD function* $\mathsf{R}_{G,X_1X_2}(D)$ *for this symmetric bivariate Gaussian source is [35, 36, Thm. III.1]:*

(i) *If* $0 < D \leq 1 - \rho$, *then*

$$\mathsf{R}_{G,X_1X_2}(D) = \frac{1}{2}\log\frac{1-\rho^2}{D^2}.$$

(ii) *If* $1 - \rho \leq D \leq 1$, *then*

$$\mathsf{R}_{G,X_1X_2}(D) = \frac{1}{2}\log\frac{1+\rho}{2D - (1-\rho)}.$$

(iii) *If* $D > 1$, *then*

$$\mathsf{R}_{G,X_1X_2}(D) = 0.$$

The worst-case RDM function shows a different behaviour on the following three regions $\mathscr{S}_1, \mathscr{S}_2, \mathscr{S}_3$, *illustrated in Figure 1.5:*

$$\mathscr{S}_1 := \left\{(D, M) : M \geq \mathsf{R}_{G,X_1X_2}(D)\right\},$$

$$\mathscr{S}_2 := \left\{(D, M) : \frac{1}{2}\log\frac{1+\rho}{1-\rho} \leq M \leq \mathsf{R}_{G,X_1X_2}(D)\right\},$$

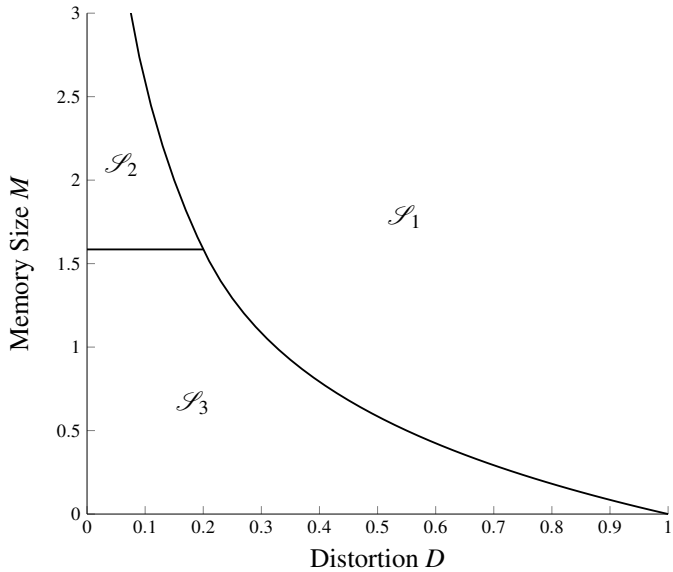

Figure 1.5: Illustration of the distortion-memory regions $\mathscr{S}_1$, $\mathscr{S}_2$, and $\mathscr{S}_3$ when $\rho = 0.8$.

and

$$\begin{aligned}\mathscr{S}_3 \quad := \quad & \left\{(D,M) : D \leq 1-\rho,\ M \leq \frac{1}{2}\log\frac{1+\rho}{1-\rho}\right\} \\ & \bigcup \left\{(D,M) : 1-\rho \leq D \leq 1,\ M \leq \mathsf{R}_{G,X_1X_2}(D)\right\}.\end{aligned}$$

In fact, the worst-case RDM function satisfies [2]:

$$\mathsf{R}_{\text{worst}}(D,D,M) = \begin{cases} 0, & (D,M) \in \mathscr{S}_1, \\ \frac{1}{4}\log\frac{1-\rho^2}{D^2} - \frac{M}{2}, & (D,M) \in \mathscr{S}_2, \end{cases}$$

and if $(D,M) \in \mathscr{S}_3$*:*

$$\mathsf{R}_{\text{worst}}(D,D,M) \leq \frac{1}{2}\log\frac{1-\frac{1}{2}(1+\rho)(1-2^{-2M})}{D}.$$

Given the symmetry of the problem, the same bounds apply also to the average-case RDM function $\mathsf{R}_{\text{avg}}(D,D,M)$*, see [4].*

Figure 1.6 illustrates the above bounds when $\rho = 0.8$*. For cache sizes* $M = \frac{1}{2}\log\frac{1+\rho}{1-\rho}$ *and* $M = \log\frac{1+\rho}{1-\rho}$ *all relevant* (D,M) *pairs are in* $\mathscr{S}_2$ *(for small cache memories* M*) or in* $\mathscr{S}_1$ *(for large cache memories* M*), and thus the exact RDM function is known. For cache size* $M = \frac{1}{4}\log\frac{1+\rho}{1-\rho}$*, the considered* (M,D) *pairs are in*

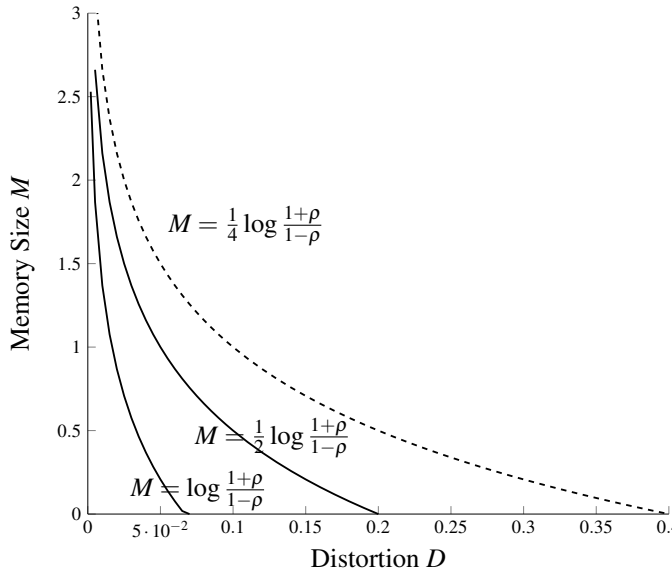

Figure 1.6: Bounds on the worst-case RDM functions for a centered bivariate Gaussian source with correlation coefficient $\rho = 0.8$ and unit variances, and for different cache sizes.

$\mathscr{S}_3$, and only an explicit lower bound on the worst-case RDM function is known. This lower bound is dashed in Figure 1.6 to emphasize that it does not necessarily indicate the exact RDM function.

1.4.2 Lower Bounds, Critical Cache Sizes, and Common Informations (CI)

As we elaborate in this subsection, the RDM function for the single-user caching setup with correlated files is also related to (lossy versions of) *Wyner's common information (CI)* and *Gác's-Körner's CI* [5–8]. This does not come as a surprise because both CIs are known for their operational meanings on the Gray-Wyner source coding network. In the Gray-Wyner network, Wyner's CI characterizes the smallest common rate R_{c} for which there exist achievable individual rates so that the sum of all rates $R_{\mathrm{c}} + R_1 + \cdots + R_L$ coincides with the joint RD-function $\mathsf{R}_{W_1\cdots W_L}(\tilde{\mathbf{D}})$ of a super-receiver that observes all the indices $J_{\mathrm{c}}, J_1, \ldots, J_L$. In the same network, the Gács-Körner CI designates the largest common rate R_{c} for which there exist achievable individual rates so that the sum of the rates $R_{\mathrm{c}} + R'_{\ell}$ to any given user ℓ' coincides with the individual RD-function $\mathsf{R}_{W_{\ell'}}(\tilde{D}_{\ell'})$ to this user.

In our single-user caching setup the lossy Wyner and Gács-Körner CIs show up when analyzing the tightness of two simple lower bounds on the worst-case RDM function.

1.4.2.1 Genie-Aided Lower Bound and Gács-Körner CI

The first lower bound is obtained by considering a related system where a genie tells the server in advance which file will be requested by the user during the delivery phase. The optimal caching strategy for this hypothetical *genie-aided* system is to compress the requested ℓ-th source W_ℓ^n using an optimal RD code for that source, to place any nM bits of the code's output in the user's cache memory, and to then send the remaining bits during the delivery phase. The RDM function of this system is easily found to be

$$\mathsf{g}(\mathbf{D},M) = \Big[\max_{\ell\in\mathscr{L}} \mathsf{R}_{W_\ell}(D_\ell) - M\Big]^+,$$

where $[a]^+$ stands for $\max\{0,a\}$. In general, the genie-aided RDM function $\mathsf{g}(\mathbf{D},M)$ provides a lower bound on the worst-case RDM function $\mathsf{R}_{\text{worst}}(\mathbf{D},M)$ of the caching system under consideration:

$$\mathsf{R}_{\text{worst}}(\mathbf{D},M) \geq \mathsf{g}(\mathbf{D},M), \qquad \forall M \geq 0. \tag{1.33}$$

Moreover, in the absence of any cache memory, i.e., for $M=0$, the lower bound is tight, $\mathsf{g}(\mathbf{D},0) = \mathsf{R}_{\text{worst}}(\mathbf{D},0)$. It can be shown that in some setups the lower bound is also tight for positive cache memory sizes $M>0$. We define the *critical cache size* M_c^* as the largest $M \geq 0$ such that $\mathsf{g}(\mathbf{D},M) = \mathsf{R}_{\text{worst}}(\mathbf{D},M)$. Since $\mathsf{g}(\mathbf{D},M)$ decreases linearly in M and $\mathsf{R}_{\text{worst}}(\mathbf{D},M)$ is convex, this implies that the genie-aided lower bound is tight for all cache sizes below M_c^*. A single-letter expression for the critical cache size M_c^* was provided in [2], where it was also shown that it is lower bounded by the L-variable lossy Gács-Körner CI:

$$\mathsf{M}_\mathsf{c}^* \geq \mathsf{K}_{\text{GK}}(W_1,\dots,W_L;D_1,\dots,D_L), \tag{1.34}$$

where equality holds whenever the single-source RD functions are all the same: $\mathsf{R}_{W_1}(D_1) = \cdots = \mathsf{R}_{W_L}(D_L)$. The lossy Gács-Körner CI $\mathsf{K}_{\text{GK}}(W_1,\dots,W_L;D_1,\dots,D_L)$ for L variables $W_1,\dots,W_L \sim P_{W_1\cdots W_L}$ is a straightforward extension of the two-variable lossy Gács-Körner CI in [7] and was formally defined in [2]:

$$\mathsf{K}_{\text{GK}}(W_1,\dots,W_L;D_1,\dots,D_L) := \max_{(U,\hat{W}_1,\dots,\hat{W}_L)} I(W_1,\dots,W_L;U), \tag{1.35}$$

where the maximum is taken over all tuples $(U,\hat{W}_1,\dots,\hat{W}_L)$ on $\mathscr{U}\times\hat{W}_1\times\cdots\hat{W}_L$ jointly distributed with $W_1,\dots,W_L$ and satisfying

(i) $\forall\, \ell' \in \mathscr{L}:\ U \leftrightarrow W_{\ell'} \leftrightarrow \{W_{\ell^c}\}_{\ell^c\in\mathscr{L}\setminus\{\ell'\}}$

(ii) $\forall\, \ell' \in \mathscr{L}:\ U \leftrightarrow \hat{W}_{\ell'} \leftrightarrow W_{\ell'}$

(iii) $\forall\, \ell' \in \mathscr{L}:\ \mathbb{E}[\delta_{\ell'}(\hat{W}_{\ell'},W_{\ell'})] \leq D_{\ell'}$

(iv) $\forall\, \ell' \in \mathscr{L}:\ I(W_{\ell'};\hat{W}_{\ell'}) = \mathsf{R}_{W_{\ell'}}(D_{\ell'})$.

When specialized to Hamming distortions and zero maximum distortions, the lossy Gács-Körner CI specializes to the original Gács-Körner CI [8] which we denote by $\mathsf{K}_{\text{GK}}(W_1,\dots,W_L)$. This latter CI can also be expressed as the largest entropy of a common random variable that can be extracted individually from each of the random

variables $W_1, \dots, W_L$. That means, the largest entropy of a random variable T for which there exist functions $f_1, \dots, f_L$ so that with probability 1:

$$T = f_1(W_1) = \cdots = f_L(W_L). \tag{1.36}$$

1.4.2.2 Super-User Lower Bound and Wyner CI

The second lower bound is obtained by the RDM function of a setup with a super-user that observes delivery messages $X_1, \dots, X_L$ corresponding to all possible L demands and, with these messages and the cache content, reconstructs the entire library $W_1^n, \dots, W_L^n$ up to the desired distortions $D_1, \dots, D_L$. The optimal caching strategy for this super-user setup is to take an optimal joint RD code that jointly describes all the files $W_1^n, \dots, W_L^n$, to then place nM bits of the code's output in the user's cache memory, and to finally distribute the remaining bits equally over the L links in the delivery phase. The RDM function $\mathsf{s}(\mathbf{D}, C)$ of this super-user problem is

$$\mathsf{s}(\mathbf{D}, M) = \left[\frac{\mathsf{R}_{W_1 \cdots W_L}(\mathbf{D}) - M}{L} \right]^+, \tag{1.37}$$

and it lower bounds the RDM function of our single-user caching problem:

$$\mathsf{R}_{\text{worst}}(\mathbf{D}, M) \geq \mathsf{s}(\mathbf{D}, M), \quad \forall M \geq 0. \tag{1.38}$$

As already mentioned in the previous section, the RDM function of our single user caching setup is trivially 0 when the cache memory is sufficiently large to store all the bits of the joint RD code:

$$\mathsf{R}_{\text{worst}}(\mathbf{D}, M) = 0, \qquad M \geq \mathsf{R}_{W_1 \cdots W_L}(\mathbf{D}). \tag{1.39}$$

Since $\mathsf{s}(\mathbf{D}, M)$ is nonnegative, the worst-case RDM trivially coincides with the super-user lower bound $\mathsf{s}(\mathbf{D}, M)$ for all cache sizes above $\mathsf{R}_{W_1 \cdots W_L}(\mathbf{D})$. It can be shown that in some problems even for smaller cache sizes the RDM functions $\mathsf{s}(\mathbf{D}, M)$ and $\mathsf{R}_{\text{worst}}(\mathbf{D}, M)$ coincide. One can thus define the *critical cache size* $\mathsf{M}_{\mathsf{s}}^*(\mathbf{D})$ as the smallest cache size where the two bounds coincide. Since the RDM function $\mathsf{s}(\mathbf{D}, M)$ is linear and the RDM function $\mathsf{R}_{\text{worst}}(\mathbf{D}, M)$ is convex, the bounds indeed coincide for all cache memory sizes above $\mathsf{M}_{\mathsf{s}}^*(\mathbf{D})$.

A single-letter expression for the critical cache size $\mathsf{M}_{\mathsf{s}}^*(\mathbf{D})$ was presented in [2], where it was also shown that it is always lower bounded by the L-variable lossy Wyner CI $\mathsf{K}_{\mathrm{W}}(W_1, \dots, W_L; D_1, \dots, D_L)$:

$$\mathsf{M}_{\mathsf{s}}^*(\mathbf{D}) \geq \mathsf{K}_{\mathrm{W}}(W_1, \dots, W_L; D_1, \dots, D_L). \tag{1.40}$$

The L-variable lossy Wyner CI of $W_1, \dots, W_L$ is defined as [7]:

$$\mathsf{K}_{\mathrm{W}}(W_1, \dots, W_L; D_1, \dots, D_L) := \min_{(U, \hat{W}_1, \dots, \hat{W}_L)} I(W_1, \dots, W_L; U),$$

where the minimum is taken over all tuples $(U, \hat{W}_1, \dots, \hat{W}_L)$ on $\mathscr{U} \times \hat{\mathscr{W}}_1 \times \cdots \times \hat{\mathscr{W}}_L$ such that the following four properties hold

(i) $(W_1, \dots, W_L) \leftrightarrow (\hat{W}_1, \dots, \hat{W}_L) \leftrightarrow U$

(ii) $\forall\, \ell' \in \mathscr{L}:\ \hat{W}_{\ell'} \leftrightarrow U \leftrightarrow \{\hat{W}_{\ell^c}\}_{\ell^c \in \mathscr{L} \setminus \{\ell'\}}$

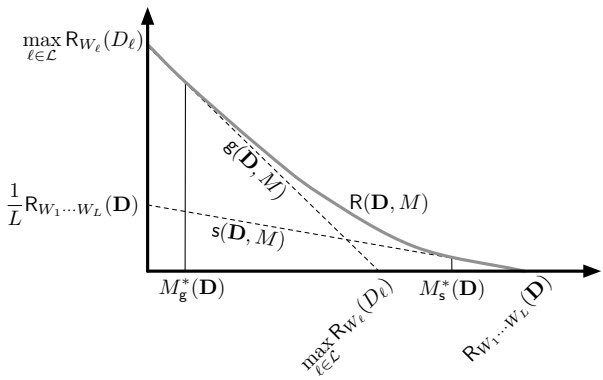

Figure 1.7: An illustration of some typical characteristics of the RDM function $\mathsf{R}_{\text{worst}}(\mathbf{D},\mathbf{M})$ for a fixed distortion tuple $\mathbf{D} = (D_1,\ldots,D_L)$.

(iii) $\forall\, \ell' \in \mathscr{L}:\ \mathbb{E}[\delta_{\ell'}(\hat{W}_{\ell'},W_{\ell'})] \leq D_{\ell'}$

(iv) $I(W_1,\ldots,W_L;\hat{W}_1,\ldots,\hat{W}_L) = \mathsf{R}_{W_1\cdots W_L}(\mathbf{D})$.

In the lossless case, i.e., under Hamming distortion measures and $\mathbf{D} = (0,\ldots,0)$, lossy Wyner CI coincides with Wyner's CI [5] that we denote $\mathsf{K}_{\text{W}}(W_1,\ldots,W_L)$. Moreover, in this case inequality (1.40) holds with equality whenever the caching problem is symmetric in the sense that $\mathsf{K}_{\text{W}}(W_1,\ldots,W_L) = I(\boldsymbol{W};U^*)$ for some U^* satisfying $H(W_1|U^*) = \cdots = H(W_L|U^*)$ and the Markov chain $W_{\ell'} \leftrightarrow U^* \leftrightarrow \{U^*_{\ell^c}\}_{\ell^c\in\mathscr{L}\setminus\{\ell'\}}$ for all $\ell' \in \mathscr{L}$.

Alternatively, the critical cache capacity $\mathsf{M}^*_{\mathsf{s}}(\mathbf{D})$ can also be defined in terms of the *average* RDM function instead of the *worst-case* RDM function. Under this average-case framework the critical cache capacity $\mathsf{M}^*_{\mathsf{s}}(\mathbf{D})$ always coincides with the lossy Wyner CI, see [3, 4].

1.4.2.3 Typical Properties of the Single-User RDM Function

Figure 1.7 provides an overview of the discussion in this subsection. It depicts the worst-case RDM function $\mathsf{R}_{\text{worst}}(\mathbf{D},M)$ of our single-user caching problem and the genie-aided and super-user lower bounds $\mathsf{g}(\mathbf{D},M)$ and $\mathsf{s}(\mathbf{D},M)$. These bounds are tight for cache sizes smaller than $\mathsf{M}^*_{\mathsf{c}}(\mathbf{D})$ or larger than $\mathsf{M}^*_{\mathsf{s}}(\mathbf{D})$, which in their turn are lower bounded by the L-variable lossy Wyner and Gács-Körner common informations, and coincide with them in some special cases. Notice in particular that the worst-case RDM function $\mathsf{R}_{\text{worst}}(\mathbf{D},M)$ decays linearly for cache sizes $M \in [0,\mathsf{M}^*_{\mathsf{c}}(\mathbf{D})] \cup [\mathsf{M}^*_{\mathsf{s}}(\mathbf{D}),\infty)$ because both lower bounds $\mathsf{g}(\mathbf{D},M)$ and $\mathsf{s}(\mathbf{D},M)$ decay linearly in the cache size M.

We can also interpret our findings for the DSBS source with zero Hamming distortions in Example 1 in the sense discussed above. For this source, the Gács-Körner CI is 0 because it is not possible to extract a common random variable individually

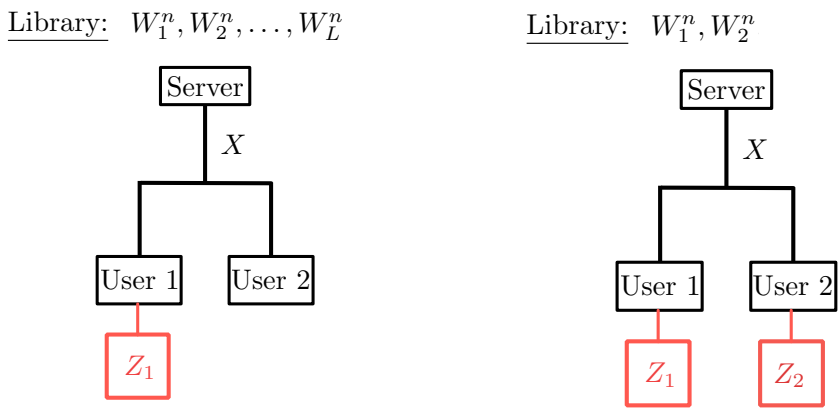

Figure 1.8: Two-user RD cache problem.

from the pair $(W_1, W_2) \sim P_{W_1 W_2}$. The standard Wyner CI for the DSBS source is well known and equals

$$\mathsf{K}_{\mathrm{W}}(W_1, W_2) = 1 + h_{\mathrm{b}}(\rho) - 2h_{\mathrm{b}}(\rho^*). \tag{1.41}$$

Since the problem is completely symmetric, the critical rate $\mathsf{M}_{\mathsf{c}}^*$ coincides with Wyner's CI $\mathsf{K}_{\mathrm{W}}(W_1, W_2)$ on this problem. In fact, as can be seen by inspecting Figure 1.4, the worst-case RDM function decays linearly for all cache sizes M above $\mathsf{K}_{\mathrm{W}}(W_1, W_2)$ where it coincides with the RDM function $\mathsf{s}(\mathbf{D})$ of the super-user caching problem.

1.5 Two-User Systems

The mulit-user caching problem is more challenging than the single-user problem. Before jumping into the $K-$user problem in the next section, we illustrate some of these challenges for the two-user problem in this section. The full solution for the two-user case is still unknown; nevertheless, lower and upper bounds and special cases are studied in [1, 11, 12].

We start this section with a simple genie-aided lower bound for the general two-user case, similar to the lower bound presented in the previous section. We then present two core caching ideas for two specific two-user systems: In Section 1.5.2, which is based on [1], we consider the scenario in Figure 1.8 (left picture) where only one user has a local cache memory, and in Section 1.5.3, which is based on [11], we consider the scenario in Figure 1.8 (right picture) with two cache memories, but only two files in the library. (A generalization to multiple files and arbitrary number of users can be found in [12].) For the two-cache scenario, the authors in [11] have focused on (almost) lossless compression. We follow their approach here; an extension to lossy reconstructions can be obtained using standard tools from RD theory.

In this section we follow the general notation introduced in Section 1.3. This means, in particular, that at user k file W_ℓ^n is measured with the per-letter distortion function $\delta_{k,\ell}$ and is subject to a maximum distortion $D_{k,\ell}$. The demand sets of the two users are denoted by $\mathscr{L}_1$ and $\mathscr{L}_2$, and their sizes by L_1 and L_2. We further use the abbreviation $\mathbf{W} := (W_1, \ldots, W_L)$ and write $\mathbf{W} \sim P_{W_1 \cdots W_L}$ to describe the underlying joint description of $\mathbf{W}$.

1.5.1 A Genie-Aided Lower Bound

Let $\mathsf{g}(\mathbf{D}, M_1, M_2)$ denote the RDM function of a two-user system with a genie that reveals the demands ℓ_1 and ℓ_2 to the server ahead of time (prior to the placement phase). For each realization of the demand pair ℓ_1, ℓ_2, where $\ell_1 \neq \ell_2$, this genie-information transforms the problem into a two-user two-file ($W_{\ell_1}^n$ and $W_{\ell_2}^n$) Gray-Wyner source coding problem where the common message J_c describes the delivery message X and the two private messages J_1 and J_2 describe the cache contents Z_1 and Z_2. The worst-case genie-aided RDM function $\mathsf{g}(\mathbf{D}, M_1, M_2)$ is then obtained from the characterization of the Gray-Wyner RD region $\mathscr{R}_\mathrm{GW}(\tilde{\mathbf{D}})$ in (1.23), and by taking the maximum over all demand pairs (ℓ_1, ℓ_2). In particular, we have

$$\mathsf{g}(\mathbf{D}, M_1, M_2) = \max_{(\ell_1,\ell_2)\in\mathscr{L}_1\times\mathscr{L}_2} \min I(U;\mathbf{W}), \tag{1.42}$$

where the minimum is over all auxiliary random variables U satisfying

$$\mathsf{R}_{W_{\ell_k}|U}(D_{k,\ell_k}) \leq M_k, \quad k \in \{1,2\}. \tag{1.43}$$

In the special case where only user 1 has a cache memory, i.e., $M_2 = 0$, the above Gray-Wyner source-coding problem reduces to the successive-refinement problem introduced and studied by Equitz and Cover [13] and by Rimold [14], and the worst-case genie-aided RDM function is directly obtained from the RD region of the successive refinement problem.[7]

$$\begin{aligned} &\mathsf{g}(\mathbf{D}, M_1, M_2 = 0) \\ &\quad = \max_{(\ell_1,\ell_2)\in\mathscr{L}_1\times\mathscr{L}_2} \min \quad \max\left\{I(\mathbf{W};\hat{W}_{2,\ell_2}),\, I(\mathbf{W};\hat{W}_{1,\ell_1},\hat{W}_{2,\ell_2}) - M_1\right\}, \end{aligned} \tag{1.44}$$

where the minimum is taken over all tuples $\{\hat{W}_{1,\ell_1}\}_{\ell_1\in\mathscr{L}_1}$ and $\{\hat{W}_{2,\ell_2}\}_{\ell_2\in\mathscr{L}_2}$ that are jointly distributed with $\mathbf{W}$ and satisfy for all $(\ell_1,\ell_2) \in \mathscr{L}_1 \times \mathscr{L}_2$:

$$\mathbb{E}\left[\delta_{k,\ell_k}(\hat{W}_{k,\ell_k}, W_{\ell_k})\right] \quad \leq \quad D_{k,\ell_k}, \quad k \in \{1,2\}. \tag{1.45}$$

Notice that the outer maximization and the minimization can be exchanged because the objective function only depends on the joint distributions of each pair $(\hat{W}_{1,\ell_1}, \hat{W}_{2,\ell_2})$ with $\mathbf{W}$.

The genie-aided RDM function $\mathsf{g}(\mathbf{D}, M_1, M_2)$ lower bounds the worst-case RDM function of our general two-user caching problem:

$$\mathbf{R}_\mathrm{worst}(\mathbf{D}, M_1, M_2) \geq \mathsf{g}(\mathbf{D}, M_1, M_2), \tag{1.46}$$

[7] The bound in (1.44) can also be obtained directly from (1.42) by rate-transfer arguments and exploiting properties of the optimal choice of U in (1.43) when $M_2 = 0$.

because genie-information can only reduce the (worst-case) RDM function. In the following two subsections, we present upper bounds on $\mathbf{R}_{\text{worst}}(\mathbf{D}, M_1, M_2)$ when $M_2 = 0$ and show special cases where they coincide with the genie-aided lower bound.

1.5.2 Two Users and One Cache

Consider the setup in Figure 1.8 (left picture). User 1 has a cache memory Z_1 with capacity M_1 bits per source symbol and user 2 does not have a cache ($M_2 = 0$). We first consider two lossy setups and then an (almost) lossless setup.

1.5.2.1 Lossy Reconstructions with $\mathscr{L}_2$ a Singleton

Consider first the special case where $\mathscr{L}_2$ is a singleton, i.e., $\mathscr{L}_2 = \{\ell_2^\star\}$ for some $\ell_2^\star \in \mathscr{L}$. In this case, the following placement and delivery strategies are optimal:

- The server quantizes $W_{\ell_2^\star}^n$ into user 2's reconstruction $\hat{W}_2^n$ such that it satisfies the desired distortion constraint $D_{2,\ell_2^\star}$. The server will send the description of $\hat{W}_2^n$ in the delivery phase as part of the message X, letting user 2 reconstruct $\hat{W}_2^n$ directly from the delivery message.
- Since user 2 can directly reproduce $\hat{W}_2^n$ from the delivery message (without help from the local cache memory), so can user 1. Given this side information (SI), the server applies an optimal placement and delivery strategy for a single-user RD caching problem where both the encoder (the server) and the decoder (user 1) have access to the SI sequence $\hat{W}_2^n$. That is, the server applies a modified Gray-Wyner source code with SI where it first quantises the entire library $W_1^n, \ldots, W_L^n$ into an auxiliary codeword U^n, assuming SI $\hat{W}_2^n$ at the encoder and decoder, and places the description of the quantized codeword into user 1's cache memory. After receiving user 1's demand ℓ_1[8], the server applies a conditional RD-code (with the target distortion $D_{1,\ell}$) to the file $W_{\ell_1}^n$ requested by user 1, assuming that SI $(U^n, \hat{W}_2^n)$ is known to the encoder and the decoder. The description of $\hat{W}_1^n$ is sent along with the description of $\hat{W}_2^n$ in the delivery phase.
- User 1 first recovers $\hat{W}_2^n$ from the delivery message X. Using $\hat{W}_2^n$ as side information, it then recovers the auxiliary codeword U^n from the cache memory. Finally, given $(U^n, \hat{W}_2^n)$ as side information and using the partial description of $\hat{W}_1^n$ via the delivery message, user 1 reconstructs $\hat{W}_1^n$.

In summary, during the placement phase, the server puts a description of codeword U^n into user 1's cache memory using SI $\hat{W}_2^n$ (which is user 2's reconstruction) and assuming that this SI is also available at the decoder. During the subsequent delivery phase, the server sends a description of $\hat{W}_2^n$ as well as a description of $\hat{W}_1^n$ using the SI pair $(U^n, \hat{W}_2^n)$. Notice that since user 1 only learns the SI $\hat{W}_2^n$ with the delivery message X, it can only make use of its cache content at the end of the delivery phase.

[8] User 2's demand has been known because $\mathscr{L}_2$ is a singleton.

For any demand $\ell_1 \in \mathscr{L}_1$, the described strategy achieves the desired distortion constraints $(D_{1,\ell_1}, D_{2,\ell_2^\star})$ if the cache size M_1 and the delivery rate R satisfy

$$M_1 > I(\mathbf{W}; U|\hat{W}_2) \tag{1.47}$$

$$R > I(\mathbf{W}; \hat{W}_2) + I(\mathbf{W}; \hat{W}_1|U, \hat{W}_2), \tag{1.48}$$

for some triple $(\hat{W}_1, \hat{W}_2, U)$ jointly distributed with $\mathbf{W}$ satisfying

$$\mathbb{E}\left[\delta_{1,\ell_1}(\hat{W}_1, W_{\ell_1})\right] \leq D_{1,\ell_1} \quad \text{and} \quad \mathbb{E}\left[\delta_{2,\ell_2^\star}(\hat{W}_2, W_{\ell_2^\star})\right] \leq D_{2,\ell_2^\star}. \tag{1.49}$$

The achieved distortions remain unchanged if instead of storing some of the bits in user 1's cache memory, the server sends them as additional bits in the delivery message to both users. This classical rate-transfer argument and the above two conditions (1.47) and (1.48) prove that the worst-case RDM function of the described setup is upper-bounded by

$$\begin{aligned} &\mathsf{R}_{\text{worst}}(\mathbf{D}, M_1, M_2 = 0) \\ &\quad \leq \min \max_{\ell_1 \in \mathscr{L}_1} \max\Big\{ I\big(\mathbf{W}; \hat{W}_{2,\ell_2^\star}\big) + I\big(\mathbf{W}; \hat{W}_{1,\ell_1}|U, \hat{W}_{2,\ell_2^\star}\big), \\ &\qquad\qquad I\big(\mathbf{W}; U, \hat{W}_{1,\ell_1}\hat{W}_{2,\ell_2^\star}\big) - M_1 \Big\}, \end{aligned} \tag{1.50}$$

where the minimum is taken over all tuples $\big(\{\hat{W}_{1,\ell_1}\}_{\ell_1 \in \mathscr{L}_1}, \hat{W}_{2,\ell_2^\star}\big)$ jointly distributed with $\mathbf{W}$ and satisfying (1.49) for all $\ell_1 \in \mathscr{L}_1$.

Using standard converse steps, it can be shown that the inequality in (1.50) is actually an equality, and the above described scheme achieves the optimal RDM function when $\mathscr{L}_2^\star$ is a sigleton.

1.5.2.2 Lossy Reconstructions and General Demand Sets

We present a general upper bound on the RDM function $\mathsf{R}_{\text{worst}}(\mathbf{D}, M_1, M_2 = 0)$ by extending the placement and delivery strategies described in the previous subsection to demand sets $\mathscr{L}_2$ that are not singletons. First recall that when $\mathscr{L}_2$ is a singleton, the server naturally knows user 2's demand *a priori* in the placement phase. It can, therefore, treat user 2's reconstruction $\hat{W}_2^n$ as SI known to both the encoder and the decoder and exploit it in designing user 1's cache content (described by the auxiliary codeword U^n).

For general demand sets $\mathscr{L}_2$, SI $\hat{W}_2^n$ is still available at user 1 to recover the codeword U^n from its cache memory. (Recall that this holds because user 2 does not have a cache memory and $\hat{W}_2^n$ must thus be entirely described by the delivery message X.) The server however does not know the SI $\hat{W}_2^n$ when placing its description of U^n in the cache memory, because it has not yet learned the demand ℓ_2. It can thus not apply a standard conditional RD code to generate U^n.

Instead, [1] proposed that the server generates the auxiliary codeword U^n so that under any possible demand $\ell_2 \in \mathscr{L}_2$ the corresponding reconstruction sequence $\hat{W}_{2,\ell_2}^n$ can serve as decoder SI. The server then places a coarse bin index for the auxiliary codeword U^n in user 1's cache memory, and refines the bin *a posteriori* after learning the demand ℓ_2. The refined bin index is sent as part of the delivery message X and is chosen to be of exactly the right size (depending on user 2's demand ℓ_2) so as to

allow user 1 to recover U^n using SI $\hat{W}^n_{\ell,2}$. This technique leads to the following upper bound on the worst-case RDM function:

$$\mathsf{R}_{\text{worst}}(\mathbf{D}, M_1) \leq \min \max_{(\ell_1,\ell_2)\in\mathscr{L}_1\times\mathscr{L}_2} \max \left\{ \begin{array}{l} I\big(\mathbf{W};\hat{W}_{2,\ell_2}\big) + I\big(\mathbf{W};\hat{W}_{1,\ell_1}|U,\hat{W}_{2,\ell_2}\big), \\ I\big(\mathbf{W};U,\hat{W}_{1,\ell_1},\hat{W}_{2,\ell_2}\big) - M_1 \end{array} \right\} \tag{1.51}$$

where the minimum is taken over all tuples $\big(U, \{\hat{W}_{1,\ell_1}\}_{\ell_1\in\mathscr{L}_1}, \{\hat{W}_{2,\ell_2}\}_{\ell_2\in\mathscr{L}_2}\big)$ that are jointly distributed with $\mathbf{W}$ and satisfy for all demand pairs $(\ell_1,\ell_2)\in\mathscr{L}_1\times\mathscr{L}_2$:

$$\mathbb{E}\left[\delta_{k,\ell_k}(\hat{W}_{k,\ell_k}, W_{\ell_k})\right] \quad \leq \quad D_{k,\ell_k}, \quad k\in\{1,2\}. \tag{1.52}$$

This bound matches the genie-aided lower bound in (1.44) in some special cases. For example, when $\mathscr{L}_1 = \{\ell_1^\star\}$ is a singleton, in which case it is optimal to set $U = \hat{W}_{1,\ell_1^\star}$.

1.5.2.3 (Almost) Lossless Reconstructions

Consider the case where both users wish to reconstruct their demanded sources $W^n_{\ell_1}$ and $W^n_{\ell_2}$ (almost) losslessy. I.e., we have

$$\delta_{k,\ell_k}(w_{\ell_k}, \hat{w}_{k,\ell_k}) = \mathbb{1}\big\{\hat{w}_{k,\ell_k} \neq w_{\ell_k}\big\}, \qquad \forall k\in\{1,2\}, \quad \forall(\ell_1,\ell_2)\in\mathscr{L}_1\times\mathscr{L}_2$$

and $\mathbf{D} = \mathbf{0}$. The bounds in (1.44) and (1.51) specialise to:

$$R_{\text{worst}}(\mathbf{0}, M_1) \geq \max_{(\ell_1,\ell_2)\in\mathscr{L}_1\times\mathscr{L}_2} \max \left\{ \begin{array}{l} H(W_{\ell_2}), \\ H(W_{\ell_1}, W_{\ell_2}) - M_1 \end{array} \right\}, \tag{1.53}$$

$$R_{\text{worst}}(\mathbf{0}, M_1) \leq \min \max_{(\ell_1,\ell_2)\in\mathscr{L}_1\times\mathscr{L}_2} \max \left\{ \begin{array}{l} H(W_{\ell_2}) + H(W_{\ell_1}|U, W_{\ell_2}), \\ H(U, W_{\ell_1}, W_{\ell_2}) - M_1 \end{array} \right\} \tag{1.54}$$

where the minimisation in (1.54) is over all auxiliaries U. The two bounds coincide for:

1. $\mathscr{L}_1 = \{\ell\}$ for some $\ell\in\mathscr{L}$ (set $U = W_\ell$); and
2. $\mathscr{L}_1 = \mathscr{L}_2 = \{\ell,\ell'\}$ for $\ell,\ell'\in\mathscr{L}$ (set $U = (W_\ell, W_{\ell'})$).

The lower bound above (1.53) is derived from the genie-aided bound in (1.46) and its tightness implies that there is no penalty due to the lack of information about ℓ_1,ℓ_2 at the server in the the placement phase.

1.5.3 Two Users and Two Files

To develop further understanding on the caching of correlated sources, we examine the lossless caching scenario illustrated in Figure 1.8 (right picture) where both users have the same demand sets:

$$\mathscr{L}_1 = \mathscr{L}_2 = \mathscr{L} = \{1,2\}, \tag{1.55}$$

and both users have identical cache sizes:

$$M_1 = M_2 = M. \tag{1.56}$$

An approximate solution to this problem can be found by a scheme that separates source coding (which is correlation aware) from coded caching (which is agnostic to correlation).

In particular, set $L = 2$ and $\mathbf{D} = 0$ in the Gray-Wyner network of Figure 1.3 and its rate-region in (1.23). The descriptions J_0, J_1, and J_2 (of rates R_c, R_1, R_2, respectively) are thus common and individual encoded messages and are designed such that W_ℓ, $\ell \in \{1,2\}$, can be recovered from (J_0, J_ℓ) almost losslessly. Once the library is encoded, the cache contents and the delivery message are designed solely based on J_0, J_1, J_2. Let $R_1 = R_2 = \tilde{R}$ such that the rate tuple $(R_c, \tilde{R}, \tilde{R})$ satisfies the inequalities in (1.23) for some choice of U with $P_{W_1|U} = P_{W_2|U}$. We will optimise over the rate tuple and the corresponding random variable U later.

The cache encoder operates as follows:

- If $M < \tilde{R}$, only the individual messages are cached (e.g. according to the scheme of [37] described in the earlier chapters).
- If $\tilde{R} \leq M < R_c + \tilde{R}$, the first $n(M - \tilde{R})$ bits of J_0 are cached at both users and the remaining cache memory is used for the caching of J_1, J_2 (e.g. according to [37]).
- If $R_c + \tilde{R} \leq M < R_c + 2\tilde{R}$, the common description J_0 is cached at both users and the remaining cache memories are used for the caching of J_1, J_2 (e.g. according to [37]).
- If $M \geq R + 2\tilde{R}$, all three descriptions can be fully stored at both users and no delivery message will be needed.

The delivery message then consists of the portion of J_0 that was not cached and a coded multicast message encoding the pair (J_1, J_2) (e.g. following [37]). Such a two-step caching scheme achieves the following upper bound on R_{worst}:

$$R_{\text{worst}}(\mathbf{0}, \mathbf{0}, M, M) \leq \begin{cases} R_0 + 2\tilde{R} - 2M \\ R_0 + \frac{3}{2}\tilde{R} - M \\ \frac{1}{2}R_0 + \tilde{R} - \frac{1}{2}M. \end{cases} \tag{1.57}$$

The above upper bound can be optimized by minimizing over U and $(R_0, \tilde{R})$. Recall that U is characterized by $P_{U|W_1W_2}$, $P_{W_1|U} = P_{W_2|U}$, and the rate tuple $(R_0, \tilde{R}, \tilde{R})$ satisfies the constraints in (1.23) for the given choice of U.

Lower bounds on the rate-distortion-memory function can also be found using the genie lower bound in (1.42)–(1.43).

$$R_{\text{worst}}(\mathbf{0}, \mathbf{0}, M, M) \geq \min \left\{ \begin{array}{l} H(W_1, W_2) - 2M, \\ \frac{1}{2}\left(H(W_1, W_2) - M\right), \\ \frac{1}{2}\left(H(W_1, W_2) + \max(H(W_1), H(W_2)) - M\right) \end{array} \right\}. \tag{1.58}$$

The RDM function can now be approximately found. In particular, the bounds in (1.57) and (1.58) coincide for the following two ranges of the cache memory size M (i) $M \leq \tilde{M}$ and (ii) $H(W_1, W_2) - 2\tilde{M} \leq M \leq H(W_1, W_2)$, where

$$\tilde{M} = \max \frac{1}{2} \min\{H(W_1|U), H(W_2|U)\},$$

and the max is over all choices of U such that $P_{W_1|U} = P_{W_2|U}$ and $W_1 \to U \to W_2$ forms a Markov chain. In other regimes of M, the gap between the upper bound in (1.57) and the lower bound in (1.58) is at most $\frac{1}{2}\min\{H(W_1|W_2), H(W_2|W_1)\} - \tilde{M}$.

1.6 K-User Systems with Uncorrelated Sources

The K-user problem is particularly challenging and we restrict to the special case where the files $W_1, \dots, W_L$ are uncorrelated, i.e., the joint probability distribution $P_{W_1 \cdots W_L}$ factorizes as:

$$P_{W_1 \cdots W_L} = P_{W_1} \cdot P_{W_2} \cdots P_{W_L}. \tag{1.59}$$

We further assume full demand sets $\mathscr{L}_1 = \cdots = \mathscr{L}_K = \mathscr{L}$ where all users potentially demand all files in the library. This setup was studied in [17–19] for Gaussian sources and squared-error distortion measures, so in this section we assume that each source distribution is

$$W_\ell \sim \mathscr{N}(0, \sigma_\ell^2), \tag{1.60}$$

for some positive variance σ_ℓ^2, and

$$\delta_{k,\ell}(\hat{w}, w) = (\hat{w} - w)^2, \quad \forall k, \ell. \tag{1.61}$$

The works in [17–19] are based on the *scalable source-coding* idea of [13–15], which we review next.

1.6.1 Scalable Source Coding

Scalable source coding, also called *successive refinement*, was introduced and studied in [13–15]. A source $W_\ell \sim P_{W_\ell}$, $\ell \in \mathscr{L}$, is called *successively refinable* with respect to a distortion measure δ_ℓ, if for any $\varepsilon > 0$ and sufficiently large blocklength n, and any $\ell \in \mathscr{L}$, positive integer ν_ℓ and decreasing sequence of distortion levels

$$\tilde{D}_{\ell,1} > \tilde{D}_{\ell,2} > \cdots > \tilde{D}_{\ell,\nu_\ell} \geq 0, \tag{1.62}$$

it is possible to encode the i.i.d. P_{W_ℓ}-source W_ℓ^n into a sequence of refinement layers

$$\bar{\mathbf{\Omega}}_\ell := (\Omega_{\ell,1}, \dots, \Omega_{\ell,\nu_\ell}) \tag{1.63}$$

of rates $r_{\ell,1}, \dots, r_{\ell,\nu_\ell}$ by means of a scalable encoder

$$\bar{\mathbf{\Omega}}_\ell = \text{sc_enc}_\ell(W_\ell) \tag{1.64}$$

so that a user that observes the first $\lambda \leq \nu_\ell$ layers $\Omega_{\ell,1}, \dots, \Omega_{\ell,\lambda}$ can produce a reconstruction

$$\hat{W}_\ell^n = \text{sc_dec}(\Omega_{\ell,1}, \dots, \Omega_{\ell,\lambda}) \tag{1.65}$$

of file W_ℓ^n that satisfies the predetermined distortion level $\tilde{D}_{\ell,\lambda}$,

$$\mathbb{E}\left[\sum_{i=1}^n \delta_\ell(\hat{W}_{\ell,i}, W_{\ell,i})\right] \leq \tilde{D}_{\ell,\lambda} + \varepsilon, \tag{1.66}$$

and so that the cumulated rate of layers $\Omega_{\ell,1},\dots,\Omega_{\ell,\lambda}$, equals the minimum rate required to attain this specific distortion level $\tilde{D}_{\ell,\lambda}$:

$$\sum_{j=1}^{\lambda} r_{\ell,j} = R_{W_\ell}(\tilde{D}_{\ell,\lambda}). \tag{1.67}$$

It is known that Gaussian sources are successively refinable under squared-error distortion measures.

1.6.2 The Layered-Approach in Caching

The works in [17–19] propose to first compress each file W_ℓ^n by means of a scalable source encoder tailored to the (reordered) permissible distortion levels $D_{1,\ell},\dots,D_{K,\ell}$, and to then design the placement and delivery schemes in a way that each user k learns sufficiently many layers of its desired file W_{ℓ_k} to meet the required distortion level D_{k,ℓ_k}. Once the set of all possibly demanded layers is identified, the problem resembles a standard coded caching problem with a library that consists of the set of all layers and where each user wishes to download multiple files (corresponding here to multiple layers).

The described setup is rather involved in general, but simplifies significantly when the maximum permissible distortion levels only depend on the user or the file but not on both. When the maximum distortions only depends on the file, i.e., $D_{k,\ell} = D_\ell$, $\forall k \in \{1,\dots,K\}$, then the setup even specializes to the standard coded caching setup because each file is encoded into a single layer. Technically more involved is the case where the maximum distortions only depend on the users but not on the files, i.e.,

$$D_{k,\ell} = D_k, \quad \forall \ell \in \{1,\dots,L\}. \tag{1.68}$$

This assumption is realistic in video streaming because video quality often depends only on the streaming terminal: a laptop requires a higher-quality video than a smaller mobile phone. In this case, (without loss of generality) we assume that the distortions are ordered:

$$D_1 \geq D_2 \geq \cdots \geq D_K. \tag{1.69}$$

Under above assumptions (1.61)–(1.69), the problem simplifies, because now the j-th layers of all files,

$$\mathbf{\Omega}^{(j)} := \{\Omega_{1,j},\dots,\Omega_{L,j}\} \tag{1.70}$$

are all of the same rate $r_{1,j} = \cdots r_{L,j}$. We can thus define the layer-j rate

$$r_j := r_{\ell',j}, \qquad \ell' \in \mathscr{L}, \tag{1.71}$$

because the j-th layers of all files are intended (if the corresponding files are requested) to the same subset of users

$$\mathscr{U}^{(j)} := \{j,\dots,K\}. \tag{1.72}$$

A natural solution is then to time- and memory-share the placement and delivery schemes of K coded caching schemes, where the j-th scheme, for $j = 1,\dots,K$, con-

siders only the users in $\mathscr{U}^{(j)}$ and the sublibrary $\mathbf{\Omega}^{(j)}$. In fact, it can be shown that this approach achieves the optimal RDM-function for two users [38]. As we will see shortly, it is however suboptimal for $K \geq 3$ users. Nevertheless, the approach leads to interesting conclusions, and we will discuss the related literature in the next two subsections.

1.6.2.1 Cache Allocation Policies and Coding Within Layers

Once the layers have been identified, one has to design a *cache-allocation policy* that at any given user assigns a given cache memory size to each layer. Two such algorithms were proposed in [17]. In the *Ordered Cache Allocation (OCA)* algorithm [17], each user k entirely stores the lowest $j_k^\star$ layers of all files in its cache memory, where the value of the index $j_k^\star$ depends on the user's cache size, it stores a part of the file-layers of layer $j_k^\star + 1$, and it stores no information about any of the higher layers. In the *Proportional Cache Allocation (PCA)* algorithm [17], any given user k allocates to each layer-j a cache memory size that is proportional to the rate of this layer r_j. Thus, denoting by $M_{k,j}$ the cache-memory rate allocated to a given layer j at a user k,

$$M_{k,j} = \frac{r_j}{\sum\limits_{j' \colon k \in \mathscr{U}^{(j')}} r_{j'}} \cdot \frac{\mathbf{M}}{L} \cdot \mathbb{1}\{k \in \mathscr{U}^{(j)}\}, \tag{1.73}$$

where $\mathbb{1}\{\cdot\}$ denotes the indicator function that equals 1 if the statement is correct and 0 otherwise. Other cache allocation strategies are of course possible. Finding the optimal allocation policy however seems difficult in general.

Once the cache allocation strategy is fixed, one has to design the coded caching algorithm for each layer $\mathbf{\Omega}^{(j)}$ and the corresponding set of users $\mathscr{U}^{(j)}$. This task is more involved than in the standard coded caching setup of Chapter 2, because typically (as the case for OCA and PCA) the cache memory allocated for a given layer j differs across users, and thus standard coded caching algorithms designed under the assumption of equal cache sizes cannot be applied directly. This problem can be circumvented by splitting each layer into sublayers, and each sublayer l is then associated the same cache memory size at a subset of users in $\mathscr{U}_l^{(j)} \subseteq \mathscr{U}^{(j)}$ and no cache memory at the remaining users of $\mathscr{U}^{(j)} \backslash \mathscr{U}_l^{(j)}$ (and thus no cache memory at all users in $\{1, \ldots, K\} \backslash \mathscr{U}_l^{(j)}$). Standard coded caching schemes as presented in Chapter 2 can then be applied to each sublayer l of layer j and to the corresponding set of users $\mathscr{U}_l^{(j)}$. For the other users in $\mathscr{U}^{(j)}$, each requested sublayer l of layer j is transmitted by simple uncoded transmission as in the absence of cache memories.

For two users and assuming Gaussian sources and squared-error distortion measures, the described approach with scalable source coding and separately coding over the layers is optimal as shown in [38]. This optimal RDM function is achieved by employing a coded caching scheme for the lower layer (intended to both users) and a simple uncoded placement/uncoded delivery scheme for the higher layer. Depending on the cache sizes uncoded or coded placement have to be combined with coded delivery for the lower layer to attain the optimal RDM function.

1.6.2.2 Coding over Layers

While appealingly simple, the proposed approach with scalable coding and separate coding over layers can be suboptimal for $K \geq 3$ users. In [19], it is proved that when one restricts to uncoded placements and linear delivery, then the delivery rate can be reduced by coding over multiple layers. The following example is taken from [19] and shows how coding across layers can be employed.

Example 3 (Example 1 in [19])**.** *Consider $K = 3$ users $1, 2,$ and 3 with cache sizes $M_1 = 0.1 \cdot L$, $M_2 = 0.2 \cdot L$, and $M_3 = 0.6 \cdot L$, where $L = |\mathscr{L}|$ denotes the number of files. As described above, a scalable encoder is applied to each file individually for the given maximum distortion levels $D_1 \geq D_2 \geq D_3$, and the j-th layer of file W_ℓ^n is denoted $\Omega_{\ell,j}$, and its rate by r_j. In this example, we assume that the maximum distortion levels $D_1 \geq D_2 \geq D_3$ are so that the layer-$1, 2, 3$ rates satisfy*

$$r_1 = 0.2, \quad r_2 = 0.1, \quad \text{and} \quad r_3 = 0.5. \tag{1.74}$$

To describe the placement phase, the lowest layer of each file is split into two sublayers of equal rates: $\Omega_{\ell',1} = (\Omega_{\ell',1}^{(1)}, \Omega_{\ell',1}^{(2)})$. The first sublayers $\{\Omega_{\ell',1}^{(1)}\}_{\ell' \in \mathscr{L}}$ are all placed at users 1 and 2 and the second sublayers $\{\Omega_{\ell',1}^{(2)}\}_{\ell' \in \mathscr{L}}$ are placed at user 3. All second layers $\{\Omega_{\ell',2}\}_{\ell' \in \mathscr{L}}$ are placed at user 2 and all third layers $\{\Omega_{\ell',3}\}_{\ell' \in \mathscr{L}}$ at user 3. It can be verified that the described placement satisfies the cache memory constraints.

During the delivery, the server sends the two signals

$$X = (\Omega_{\ell_3,1}^{(1)} \oplus \Omega_{\ell_1,1}^{(2)}\ ,\ \Omega_{\ell_3,2} \oplus \Omega_{\ell_2,1}^{(2)}), \tag{1.75}$$

where $\oplus$ denotes the componentwise XOR operation between the bits of the corresponding (sub)-layers. With this delivery signal and its own cache content, user 1 can recover the missing sublayer $\Omega_{\ell_1,1}^{(2)}$, and combined with sublayer $\Omega_{\ell_1,1}^{(1)}$ in its cache memory, it can reconstruct $W_{\ell_1}^n$ with the desired distortion D_1. User 2 recovers sublayer $\Omega_{\ell_2,1}^{(2)}$ and with $\Omega_{\ell_2,1}^{(1)}$ and $\Omega_{\ell_2,2}$ from its cache memory it can reconstruct $W_{\ell_2}^n$ with distortion D_2. Finally, user 3 recovers both $\Omega_{\ell_3,1}^{(1)}$ and $\Omega_{\ell_3,2}$ from the delivery signal, and with $\Omega_{\ell_3,1}^{(2)}$ and $\Omega_{\ell_3,3}$ in its cache memory it can reconstruct its desired file $W_{\ell_3}^n$ up to distortion D_3.

The delivery-rate in the proposed scheme is 0.2. If one insists on coding over each layer separately, then under any uncoded placement and linear delivery strategy, the delivery rate cannot be smaller than 0.2167 [19].

1.6.3 Optimal Cache Allocation and Uncoded Placement – Linear Delivery Schemes

As previously mentioned, it seems challenging to find the optimal cache-allocation to the different layers in general. The problem becomes tractable when restricting to uncoded placement and linear delivery schemes, where delivery can code across layers. In fact, [19] showed that under these assumptions, even when one allows

for coding over the layers (during the delivery phase), the cache allocation problem turns into a linear programming problem (LP).

Currently there are thus two approaches for the K-user Gaussian RD-problem with uncorrelated sources, both based on scalable coding with layers. The approach in [19] restricts to uncoded placement and linear delivery, but optimizes over the cache allocation to the various layers. The approach in [17] fixes the cache allocation algorithms (PCA or OCA) and depending on the cache size employs uncoded or coded placement. Which of these two approaches performs better depends on the cache sizes at the users.

1.6.4 Cache Allocation Across Users

Reference [17] also considered the cache allocation problem across users, where the total cache memory budget M_{tot} is fixed and one optimizes the cache allocation to the different layers *and users*.[9] The optimization is again performed under the assumption of uncoded placement and coded delivery, possibly across layers. More specifically, an LP optimization problem is phrased to identify the optimal cache allocation to users and layers as well as the optimal linear delivery scheme. It is proved that without loss in optimality[10] delivery can be performed separately for each layer and that if the total cache budget M_{tot} permits, the optimal cache allocation assigns cache size

$$M_{k,j}^{(\text{LP})} = \frac{t_j r_j}{K-j+1} \cdot \mathbb{1}\{k \in \mathscr{U}^{(j)}\}, \qquad k \in \mathscr{U}^{(j)}, \tag{1.76}$$

to layer j and user k, for some decreasing sequence of positive integers $t_1, \dots, t_K$ for which the cache allocation in (1.76) respects the total cache memory budget:

$$\sum_{k=1}^{K} \sum_{j=1}^{K} M_{k,j}^{(\text{LP})} = M_{\text{tot}}. \tag{1.77}$$

Notice that the cache allocation strategy in (1.76) allocates the same cache size for a given layer j to all users $k \in \mathscr{U}^{(j)}$, and therefore the standard coded caching schemes described in Chapter 2 are optimal on each layer. For cache memory budgets M_{tot} where no appropriate sequence of positive integers $t_1, \dots, t_K$ exists, the optimal cache allocation is obtained by time- and memory-sharing two appropriate cache allocations as in (1.76).

1.6.5 Minimizing the Average Distortions

A similar setup to [17] and [19] was considered in [18], but with the goal to minimize the average *expected* distortion over all sources, all demands (which follow a given popularity distribution), and all users under a given constraint on the *expected* total delivery rate. The previously explained scalable coding approach is used, and

[9] The cache-allocation-across-users problem was also considered for the standard lossless coded caching setup with uncorrelated files when delivery is over noisy broadcast channels [20, 21].

[10] This holds under the total cache memory budget constraint, but not under individual cache size constraints.

uncoded placement and uncoded or coded delivery schemes are designed. Coding is even allowed over layers, as in [19].

For purely uncoded delivery, [18] proposes a water-filling cache allocation algorithm where user k allocates cache size

$$M_{k,j} = \max\left\{0,\ \frac{1}{2}\log_2\left(\sqrt{\frac{2\ln(2p_{k,j}\sigma_j^2)}{\lambda_k^*}}\right)\right\}, \tag{1.78}$$

to layer j, where recall that $p_{k,j}$ denotes the probability that user k wishes to download file W_j, and λ_k^* is a positive normalization parameter that ensures the constraint on the total cache size:

$$\sum_{j=1}^{L} M_{k,j} = M_k, \quad j \in \{1,\ldots,K\}. \tag{1.79}$$

A smaller expected distortion can be achieved by a hybrid uncoded/coded delivery scheme, where files are split into two parts and uncoded delivery is applied to the first part and coded delivery to the second part. More specifically, for the coded delivery scheme, [18] proposes to extend *Random Aggregate Popularity caching with Greedy Constrained Coloring (RAP-GCC)*, a decentralized caching algorithm proposed in [39], to account for the variable cache sizes across users and for the fact that different users request a different number of layers. The main novelty in the new algorithm, termed *Greedy Constrained Coloring (RF-GCC)* algorithm, is that users randomly place bits from the different layers that they are interested in, and the random conflict graph describing the delivery is designed across all layers.

1.7 Summary and Conclusions

We introduced a rate-distortion (RD) framework for lossy caching to quantify trade-offs between delivery rate, distortion, and cache memory sizes. In the special case where the various files are independent of each other, a layered approach based on scalable source coding is proposed. In this approach, each file W_ℓ^n is represented using a set of layers in a way that a user k can reconstruct this ℓ-th file with the desired distortion $D_{k,\ell}$ when learning the first $j_{k,\ell}$ layers. Different strategies have been proposed to place and deliver the various layers to the users, including specific cache allocation algorithms and delivery schemes that code within layers or across layers. Finding the optimal strategy remains an open problem.

In the case where the files in the library are correlated, "common parts" of the various files should be placed in the cache memories of the users, because they will be useful under various demands. This can be made more precise for the single-user caching system. Timo *et al.* showed that under certain symmetry assumptions and for relatively small cache sizes, it is optimal to place the lossy Gàcs-Körner common part in the cache memory. For sufficiently large cache sizes, they showed that it is optimal to place the lossy Wyner common part. These observations can also be rephrased in terms of common informations (CI). The lossy Gács-Körner CI characterizes the largest cache size for which there is no penalty incurred because

the server learns the demands only *after* the placement phase. Similarly, the lossy Wyner CI characterizes the smallest cache size where the cache content is *jointly* maximally useful for all possible demands.

The general rate-distortion caching problem seems hard. For single-user systems, exploiting connections to the lossy Gray-Wyner source coding network, it is possible to characterise the rate-distortion-memory (RDM) function. In the context of two users, two specific examples have been considered in the literature: one were only the first user has a cache memory and one where the library contains only two files. For the latter problem, Hassanzadeh *et al.* proposed to perform Gray-Wyner source coding and coded caching separately, and they showed that this approach attains near-optimal RDM tradeoffs. For the setup with a single cache memory, Timo *et al.* proposed a joint architecture, where in particular the delivery information addressed to the cache-free user can be used by the other user to decode its cache content. The proposed scheme achieves the optimal RDM function in some special cases, and it applies adaptive binning and rate-transfer from the placement to the delivery phase.

To summarize, the RD caching problem embraces both correlated files and lossy reconstructions, and remains widely open, especially for larger number of users. Solutions were found in interesting special cases. The main technical ingredients used in RD caching are lossy Gray-Wyner source coding, lossy Wyner CI and lossy Gács-Körner CI, joint source coding and caching, adaptive binning and rate-transfer, scalable coding, and coding within and across layers.

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